

MAKOANYE MAPHUTSENG

Soil Scientist | GHG Inventory Specialist | Environmental Data Analyst

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PROFESSIONAL SUMMARY

Environmental modeller and soil scientist with expertise in land-use systems, climate change impacts, geospatial analysis, and national greenhouse gas inventories. Over five years of professional experience supporting environmental assessment, climate reporting, and sustainable resource management through advanced data analysis and modelling.

Experienced in integrating land-use, climate, soil, hydrological, and remote sensing datasets to support evidence-based decision-making. Skilled in SWAT hydrological modelling, CMIP6 climate scenario analysis, XGBoost machine learning, GIS, remote sensing, Google Earth Engine, R, and Python. Proven experience coordinating AFOLU/FOLU greenhouse gas inventories in accordance with the 2006 IPCC Guidelines and supporting national reporting under the UNFCCC Enhanced Transparency Framework.

Interested in sustainable land-use planning, environmental modelling, carbon dynamics, biodiversity conservation, and climate adaptation research.

EDUCATION

Master of Science – Integrated Catchment and Water Resources Management | 2025

National University of Lesotho (NUL)

Dissertation: *Dissertation: Assessment of Climate Change Impacts on Streamflow Patterns and Hydrological Extremes in the South Phuthiatsana Catchment Using Hydrological Modelling Under SSP2-4.5 and SSP5-8.5 Scenarios.*

- Developed, calibrated, and validated a SWAT+ hydrological model for streamflow simulation.
- Integrated CMIP6 climate projections under SSP2-4.5 and SSP5-8.5 scenarios to assess future hydrological responses.
- Applied XGBoost machine-learning techniques to analyse hydroclimatic drivers and improve predictive understanding of hydrological extremes.
- Conducted statistical trend analysis, model evaluation, and climate impact assessment using R and Python.
- Generated evidence to support climate adaptation and integrated water resources management planning.

Bachelor of Science in Agriculture (Soil Science) | 2014

National University of Lesotho (NUL)

PROFESSIONAL EXPERIENCE

FOLU Sector Technical Contributor – 5th National GHG Inventory • Lesotho Meteorological Services
(BTR1/NC4 Project) | *March 2024 – Present (Part-time)*

- Compile and process activity data and emission factors for the Forestry and Other Land Use (FOLU) sector in line with 2006 IPCC Guidelines for National GHG Inventories.
- Conduct quality control checks ensuring consistency with prior inventory submissions, improving transparency and accuracy of national LULUCF reporting.
- Apply technical insights from role as AFOLU sector coordinator during the 4th GHG Inventory cycle to strengthen methodological continuity.
- Support the development of Lesotho's Biennial Transparency Report in compliance with UNFCCC Enhanced Transparency Framework requirements.

Project Assistant • Lesotho Meteorological Services | February 2022 – January 2024

- **Led data collection, processing, and quality assurance for Agriculture, Forestry and Other Land Uses sector, directly contributing to Lesotho's national climate reporting under the UNFCCC.**
AFOLU Coordinator – 4th National GHG Inventory:
- Designed and implemented secure data management and archiving procedures; applied rigorous QA/QC protocols to meteorological and land use datasets.
- Produced and disseminated Agrometeorological bulletins and seasonal forecasts to public and stakeholders across selected constituencies.
- Represented the Lesotho Meteorological Services in the Lesotho Vulnerability Assessment Committee (LVAC), contributing to food security monitoring and drought impact analysis.
- Provided technical support to government ministries and civil society organisations on using climate data for evidence-based decision-making.
- Distributed seasonal climate performance results through stakeholder meetings across three constituencies in Molepolole's Hoek district.

Agricultural Assistant • Ministry of Agriculture, Food Security, and Nutrition | April 2018 – March 2021

- Supported farmers in adopting climate-resilient and sustainable land management practices through training, knowledge transfer, and exposure to new agricultural inputs and processes.
- Conducted scheduled field monitoring and data collection on agricultural production at Likotsi sub-centre.
- Provided farmers with market information and connected them to viable outlets for their produce.
- Participated in government mass animal vaccination campaigns for anthrax control.

CERTIFICATIONS & TRAINING

UNFCCC Technical Expert – Biennial Transparency Reports

United Nations Climate Change Secretariat, 2024. Training programme for technical experts participating in the technical expert review of biennial transparency reports.

Certificate of Proficiency – Forestry and Other Land Uses (FOLU)

GHG Management Institute (GHGMI), 2021. Based on 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Cloud-based Earth Observation for Surface Water Monitoring

Waternet International Summer School / EO Africa R&D Facility, Maseru, 2024. Face-to-face training in cloud-based EO image analysis for monitoring surface water dynamics using Google Earth Engine.

Water Pollution Management in Achieving SDG 6.3

Cap-Net UNDP, 2025. Comprehensive training on water pollution drivers, prevention strategies, stakeholder engagement, and policy integration.

KEY PROJECTS

National GHG Inventory – AFOLU Sector Coordination (4th Cycle)

Led end-to-end data collection, analysis, and quality control for the Agriculture, Forestry and Other Land Use sector during Lesotho's 4th National GHG Inventory. Demonstrated capacity for large-scale environmental data processing and assurance in compliance with international reporting standards.

MSc Dissertation – Climate Change and Hydrological Extremes

Analysed historical and projected climate data to assess streamflow variability and hydrological extremes in the South Phuthiatsana Catchment. Applied hydrological modelling and statistical scenario analysis to support integrated water resources management and adaptation planning.

Lesotho Vulnerability Assessment and Analysis (2020–2021)

Contributed timely data analysis to the national vulnerability assessment, integrating nutrition, gender, and chronic illness dimensions into drought impact evaluations to inform government interventions across food security, water, and health sectors.

MODELING AND ANALYTICAL METHODS

- Hydrological Modeling: SWAT+
- Machine Learning: XGBoost
- Climate Change Impact Assessment
- CMIP6 Climate Model Analysis
- Scenario Analysis (SSP2-4.5 and SSP5-8.5)
- Statistical Modeling and Time-Series Analysis
- Spatial Analysis and GIS
- Remote Sensing and Google Earth Engine
- Environmental Data Management and QA/QC
- Carbon Accounting and FOLU/AFOLU Inventories

TECHNICAL SKILLS

GHG Inventory & Climate Policy: 2006 IPCC Guidelines, FOLU/AFOLU sector compilation, QA/QC for national inventories, UNFCCC Enhanced Transparency Framework, Biennial Transparency Reports

Geospatial & Remote Sensing: Advanced GIS (ArcGIS, QGIS), Google Earth Engine, Remote Sensing for land cover and surface water monitoring

Programming & Data Analysis: R, Python, SPSS, R-Instat, Climsoft; statistical modelling, hydrological simulation, data cleaning and QA/QC

Domain Knowledge: Soil Science, Integrated Water Resources Management, Climate Change Adaptation, Agronomy, Horticulture, Environmental Management, Sustainable Land Use, Food Security

Soft Skills: Stakeholder engagement, technical report writing, cross-institutional collaboration, self-directed learning, adaptability

REFERENCES

Khoboso Seutloali

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Kabelo Lebohang

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